

UbiComp4All: 1st International Symposium on Inclusive and Equitable Ubiquitous Computing

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ABSTRACT

We propose a symposium to foster research on inclusive and equitable ubiquitous computing in Latin America (LATAM). The meeting is inspired by the ACM SIGCHI Across Borders Initiative, and the 1st CSCW@LatAm Research Catalyst Workshop. We aim to: 1) discuss opportunities and challenges of defining and accomplishing inclusive and equitable ubiquitous computing centered on LATAM; 2) collaboratively mentor emerging related projects focused on LATAM; 3) create a space that encourages cross-country collaborations to address together common research challenges, whilst also strengthening the links among all the members, regardless of their level of expertise in the field; and, 4) provide a forum for dialogue between local communities in Latin America (e.g., educational and health workers, NGOs) and UbiComp researchers, with the aim of identifying pathways for codesign-focused collaborations between diverse groups of stakeholders on problems of public interest.

CCS CONCEPTS

• **Human-centered computing** → *Ubiquitous and mobile computing*; **Human computer interaction (HCI)**.

KEYWORDS

Inclusive ubiquitous computing, equitable computing, LATAM, accessibility, community, social good

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1 INTRODUCTION

Despite tremendous progress made by the proponents of ubiquitous computing over the last several decades, its benefits are yet to reach communities around the world in an equitable way. In particular, several regions of the world, including countries in the Global South (e.g., in Latin America, Africa, South-East Asia) remain underserved by ubiquitous computing. Even in the Global North (e.g., the US), the penetration and impact of ubiquitous computing have not been equitable across social groups. This trend is also reflected in the participation at conferences on ubiquitous computing, as well as in the technology industry more generally. Although the latter has aimed to address the issue by adapting/deploying products for underserved communities, this has raised ethical concerns about techno-colonialism [1] and generated discontent among target user groups. We believe that the vision of achieving equity in computing—across geographically, socially, or economically underserved groups—requires strategic planning through engagement with various stakeholder audiences to constructively operate within the constraints of their needs and resources.

Furthermore, inclusive ubiquitous computing refers to the design and deployment of ubiquitous computing devices (from portable computers to technologically endowed objects such as glasses and clothing) in ways that consider differing user capabilities. Multi-disciplinary research is needed in order to address both social and technical issues in this realm. Latin American countries exhibit traits and pose challenges that merit the development of specific approaches and generate unique collaboration opportunities. The proposed symposium attends to the need to further explore such approaches and opportunities.

2 UBICOMP RESEARCH AND PRACTICE IN LATIN AMERICA

UbiComp and HCI research have followed a strong trend grounded in socio-cultural aspects from the Global North. However, as research activity in these fields becomes more distributed around the world, there has been a growing need to deepen the understanding

of underserved communities and their interactions with computing technologies. In the case of Latin America (LATAM), UbiComp research has been conducted in countries like Mexico and Brazil; however, their findings have not yet received much exposure in more local settings. Similar to other areas within HCI, the relative invisibility of UbiComp in this region may be explained by the visible gap between the dominant narratives around global and local impact; language barriers, lack of research infrastructure, among others contribute to this gap [5, 7].

LATAM has a unique set of social, cultural, and technical characteristics; therefore, it also has its own set of societal challenges that could benefit from technology. On one hand, there are challenges related to perspectives on personal data, ethics, and privacy in the region, as well as factoring of cultural sensitivities in the design of new technologies. On the other hand, there is a need for systems and services that address issues prominent in LATAM communities such as population, migration, aging, crime, disability, and diversity. For example, the National Council of Science and Technology in Mexico in its Call for National Strategic Programs states that one of the urgent challenges in the country is pollution, based upon severe air quality problems, as well as toxic levels in their waters at certain regions of the country. Likewise, another challenge that this organization urges to work on, is related to the early detection of different types of cancer [6]. Finally, migration is a very important issue not only for Mexico but for several LATAM countries, and involves factors such as drug cartels, violence, poverty, forced displacement, among others [2].

The UbiComp research community alone cannot solve these challenges, but we can begin to engage in dialogue with diverse stakeholder groups in ways that enable research solutions in these domains, for example by gathering relevant data using ubicomp devices and deriving actionable insights towards engaging policy makers among other stakeholders. For this to happen, it is essential that we identify pathways for sustained engagement with local communities and collaboratively design and develop solutions to aforementioned problems of public interest.

3 MOTIVATION AND PREVIOUS WORK

On one hand, this symposium has a case study as a starting point for discussion, which was modeled after the 1st CSCW@LatAm Research Catalyst Workshop, held in 2019 in Panama City, Panama. The goal of that event was to expedite the formation of collaborative multi-year research agendas among existing CSCW LatAm research groups [4]. Among those groups, a team of researchers currently based in Brazil, Chile, Mexico, and Panama, under the mentoring of Prof. Myriam Lewkowicz, envisioned a multi-national project titled "Building LatAm Ecosystems for Supporting Vulnerable People". The goal of this project is to create a portfolio of design case studies in LATAM countries that will support various stakeholders in the application domain of inclusive and equitable ubiquitous computing to facilitate collaboration and create synergies to address common challenges. In the final session of the CSCW@LatAm workshop, consensus was reached regarding the need to continue work in a distributed manner, as well as the advantages of holding a follow-up workshop, in which ongoing research results could be shared and discussed.

On the other hand, we want to provide a forum for interaction between local communities in Latin America (consisting of teachers and health workers, among others) and UbiComp researchers located in Latin America and other parts of the world, with the goal of fostering collaboration among these stakeholders on problems related to the public interest. The organizers have been working towards fostering HCI research activity in Mexico and LATAM for several years.

4 SYMPOSIUM OBJECTIVES AND OUTCOMES

Our symposium aims to provide a forum for interaction between local communities in Latin America (consisting of educational and health workers, immigrants, people with disabilities (PwD), elderly persons, among others) and UbiComp researchers located in Latin America and other parts of the world, with the goal of fostering collaboration among these stakeholders on problems related to the public interest. The symposium program will alternate between short presentations by the participants and breakout sessions that will focus on challenges and opportunities for UbiComp research in the LATAM context that delve into the areas discussed. In particular, we will focus on problems of public interest that are aligned with interests of representatives from local communities attending the symposium (e.g., educational and health workers, PwD). Moreover, we will explore ways for UbiComp technologies to be suitably integrated in the target communities, rather than merely being deployed for small-scale studies.

We want to create an engaging space for an open discussion of research in inclusive and equitable ubiquitous computing research from the LATAM perspective: its achievements, opportunities, challenges, commonalities and differences with other regions.

In order to reach our objective, during this symposium, we plan:

- To discuss the opportunities and challenges of defining and accomplishing inclusive and equitable ubiquitous computing research centered on LATAM
- To collaboratively mentor emerging projects focused on LATAM
- To create a space that encourages cross-country collaborations to address together common research challenges, while also strengthening the ties between the more experienced and junior members of the community
- To provide a forum for dialogue between local communities in Latin America (e.g., educational and health workers, NGOs) and UbiComp researchers, with the aim of identifying pathways for codesign-focused collaborations between diverse groups of stakeholders on problems of public interest.

5 CALL FOR CONTRIBUTIONS

The symposium will have two submission tracks:

5.1 Track 1: Inclusive Ubiquitous Computing

This track has a special emphasis on exploring the design of UbiComp technologies for marginalized communities and users with special needs in Latin America (LATAM). We solicit research papers (up to 4 pages including references). Topics of interest include, but are not limited to:

- Qualitative or quantitative inclusive ubiquitous computing studies in or about Latin America.
- Inclusive ubiquitous computing for special needs in LATAM context.
- Inclusive ubiquitous computing centered around vulnerable and marginalized people and communities in LATAM.
- Inclusive ubiquitous computing in education, health, among other areas.

5.2 Track 2: Equitable Ubiquitous Computing

This track aims to foster dialogue between local communities in Latin America (e.g., teachers, health workers, NGOs) and UbiComp researchers, with the aim of identifying pathways for codesign-focused collaborations between diverse groups of stakeholders on problems of public interest. We solicit two categories of contributions:

Research or technical papers (up to 4 pages including references) should report an ongoing or recently finished research that involves the use of ubiquitous computing in solving problems of public interest or social good. Topics of interest include, but are not limited to:

- UbiComp in Specific Application Areas, such as Public/Global Health, Education, Agriculture, Refugee Migration.
- Systems and infrastructures that support ubiquitous computing for social good.
- Empirical study of applications that leverage UbiComp devices and systems for social good.
- Methodologies and tools applied to studying UbiComp applications.
- AI and other data-driven approaches to building UbiComp technologies for social good.

Problem abstracts (up to 2 pages including references) that describe an important social good challenge relevant for a chosen social group or community in LATAM. We anticipate that these contributions will come from NGOs (e.g., environmental groups or health agencies) and will not require them to present any UbiComp-based solution to the problem. The key objective here is to make UbiComp researchers aware about possible problems on which they can collaborate with the local stakeholders. To minimize language barriers, we will allow the problem abstracts to be submitted in Spanish or Portuguese, and translate the accepted abstracts to English for benefit of the attendees and readers.

5.3 Publication

Papers and problem abstracts of both tracks should use the ACM SIGCHI Master Article template with 2 columns. The accepted papers are to be included in the ACM Digital Library. However, for the symposium website, we will make all papers available in any format of choice for the participants. At least two reviewers from our organizer committee and PC members will peer-review each submission. We expect to have a selection of 10-20 accepted papers. Given the theme of this symposium, we are confident to attract a large number of paper submissions, which will result in a high-quality symposium.

5.4 Symposium Timeline

- Submission Deadline: May 26, 2023
- Notification: June 30, 2023
- Camera-ready Deadline: July 31, 2023

5.5 Symposium publicity

The symposium will be advertised via SIGCHI, SIGMobile, and UbiComp, Hispanics in Computing and LAIHC mailing lists, social media channels, as well as news and media outlets within Mexico and internationally (e.g., the HCI Across Borders Facebook group). We will also leverage our connections with academic institutions and local grassroots communities in Mexico. Finally, we will advertise the symposium through communities and ACM chapters based in Latin America.

6 AGENDA

In the 1-day symposium, we will bring people together from academia, industry and NGO who are active within areas related to inclusive and equitable ubiquitous computing in LATAM. We plan to start the symposium with a keynote speech by a luminary in the field of UbiComp or HCI4D research for both tracks.

For Track 1, we structured different sessions such as: introduction (cards + invited talks), affinity diagrams and posters, collaborative mentoring tables, and posters and closing. For Track 2, after keynote, there will be panels (where each panel will consist of a group of short presentations by participants who made symposium submissions, arranged according to themes) and breakout sessions that will focus on challenges and opportunities for UbiComp research in the LATAM context.

The organizers also plan to invite at least 4-5 senior UbiComp researchers to the symposium— including those who have served on the Program Committee of UbiComp and Editorial Board of IMWUT—to provide feedback and mentorship, but more importantly to help brainstorm models and mechanisms for moving forward given our stated goals. The discussions and outcomes generated during the symposium will serve as the framework for follow-on encounters, workshops, and developments, following the same strategy that other LATAM initiatives [3] [5] [4]. As mentioned previously, in our experience, previous integration efforts have lead to an increase in participation from LATAM UbiComp researchers.

7 ORGANIZERS

Laura S. Gaytán-Lugo is a research professor at the Universidad de Colima in Mexico. She focuses on HCI. She is a member of the National System of Researchers in Mexico and the Mexican Academy of Computing. She is co-coordinator of the SIGCHI Latin America Committee (SLAC).

Pedro C. Santana-Mancilla is a professor and researcher at the Universidad de Colima in Mexico. His research interests are focused on HCI, IA, IoT, and Technologies to support older adults and learning. He coordinates the HCI section of the Mexican Academy of Computer Science (AMexComp). He serves as the president of the Mexican Association on Human-Computer Interaction (AMexIHC).

Soraia Prietch is a Professor at the Federal University of Rondônia, Brazil. Her research focuses on User Experience, Assistive Technology and Educational Technology. Currently, she is a

postdoctoral researcher at BUAP, México, and serves as Research Community Vice Chair of the ACM SIGCHI Latin American HCI Community (LAIHC).

Francisco J. Gutiérrez is an Assistant Professor in Human-Centered Computing at the University of Chile. His current research interests are the design, use, and evaluation of interactive systems in extended reality environments, particularly aiming to support presence, involvement, and immersion in several application domains, such as healthcare, education, gaming, and assistive technologies. He is also interested in studying the human and social factors in the interaction within the so-called metaverse, including the moral and ethical dilemmas that lie in the intersection between the user, society, and technology. He currently chairs the Santiago-Chile ACM SIGCHI Chapter and acts as the Chile liaison for the ACM SIGCHI Latin American Committee.

J. Alfredo Sánchez is a researcher of the National Laboratory of Advanced Informatics (LANIA) in Mexico. He co-founded the Latin American Conference on Human-Computer Interaction (CLIHC), the Mexican Conference on Human-Computer Interaction (Mex-IHC) and the SIGCHI Latin American Community (LAIHC). His research interests include assistive technologies, user experience design for deaf users and natural user interfaces.

Cuauhtémoc Rivera-Loaiza is an Assistant Professor at the School of Physics and Mathematics of the Universidad Michoacana

de San Nicolás de Hidalgo, in Morelia, Mexico. His research is centered in the field of Human-Computer Interaction. Most recently, he has directed his research efforts to the migrant population in Mexico and the USA. He is a founding member of CHI-Mexico and AMexIHC and was the general co-chair for MexIHC 2022. Since 2023, he has been a member of the SIGCHI Global Committee.

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